Real Estate Price Exploration Tool

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Declaration

The Project Report entitled **"Real Estate Price Exploration Tool"** is a record of bonafide work of **N. NITHIN (2420030488), S. LOCHAN (2420030413), K. SAI PRANAV (2420030224)** under the guidance of **[MS.Prathusha]**, [Designation], Department of Computer Science and Engineering, KLH CSE Bowrampet Campus, submitted in partial fulfillment for the award of B.Tech to the K L (Deemed to be) University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled **"Real Estate Price Exploration Tool"** is a bonafide work done and submitted by **N. NITHIN (2420030488), S. LOCHAN (2420030413), K. SAI PRANAV (2420030224)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of CSE, K L (Deemed to be University), during the academic year **2026–2027.**

Signature of the Guide

Signature of the Course Coordinator **Signature of the HOD**

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Abstract

Buying, selling or renting a home is one of the largest financial decisions most people make, yet property prices are hard to compare. Prices depend on location, built-up area, number of bedrooms and bathrooms, age of the property, amenities and market conditions, and listings are usually scattered across portals and spreadsheets without any consistent way to explore them.

The project, “Real Estate Price Exploration Tool”, proposes an interactive tool that lets users explore property price data through filters, summary statistics and visualizations. The workflow covers data collection, cleaning and handling of missing values, exploratory data analysis (EDA), feature encoding, and comparison of Linear Regression, Random Forest Regression and Gradient Boosting Regression for estimating the price of a property from its attributes. The results are presented in a dashboard where users can compare localities, view price-per-square-foot trends, study the relationship between area and price, and obtain an indicative price estimate.

The tool is intended as a research and planning aid. Estimates are model-based approximations drawn from the chosen dataset; they are not a valuation, a property appraisal or financial advice. Model accuracy figures are not claimed in this report and must be recorded from real experiments once the system is built.

Keywords: Real estate, house price prediction, exploratory data analysis, machine learning, Random Forest, Gradient Boosting, data visualization, Streamlit.

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CHAPTER 1
INTRODUCTION

1.1 Background of the Project

The real estate market is influenced by many interacting factors: the locality, the size and condition of the property, nearby facilities, infrastructure development and overall demand. Buyers, sellers and tenants often rely on word of mouth or on listing portals that show individual properties but give little insight into how prices vary across an area.

Data analysis offers a systematic alternative. By collecting property records into a structured dataset, cleaning them and exploring them with statistics and charts, hidden patterns such as price differences between localities or the effect of area on price become visible. Real Estate Price Exploration Tool is a proposed application that combines exploratory data analysis, interactive visualizations and a machine learning based price estimator in one dashboard.

1.2 Problem Statement

Property price information is fragmented and difficult to compare. A single listing does not show whether the asking price is high or low for the locality, how price per square foot changes with location or property age, or which features contribute most to price. Manual comparison using spreadsheets is slow, repetitive and error-prone. The project addresses this by providing one interface to filter, analyze, visualize and estimate property prices from a cleaned dataset.

1.3 Scope of the Project

The proposed system includes loading and cleaning a property price dataset; interactive filters for location, price range, area, bedrooms and property type; summary statistics and EDA outputs such as distributions, missing-value summaries and correlations; comparison of localities; and a price estimation module trained on the dataset. The system supports decision-making but does not replace professional valuation, legal verification or advice from a registered property consultant.

1.4 Objectives

1. Collect and clean a real estate dataset so that it is reliable for analysis.
2. Provide interactive charts for price distribution, price per square foot, locality comparison and area-versus-price relationships.
3. Allow users to search and filter properties by location, budget, size and other attributes.
4. Compare regression models and use the best-performing one to give an indicative price estimate.
5. Present findings in a simple dashboard that non-technical users can understand.

1.5 Importance of the Application

The tool reduces the effort needed to understand a property market. Buyers can check whether a price is reasonable for the area, sellers can position their listings realistically, and students and analysts can study how different features influence price. Transparent visualizations make the reasoning behind the numbers easy to inspect.

1.6 Target Users / Audience

- Home buyers and tenants comparing localities and budgets.
- Property owners and sellers estimating a reasonable listing price.
- Real estate agents and analysts reviewing market trends.
- Students and researchers studying housing data and price prediction.

CHAPTER 2
SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Component	Suggested Requirement
Computer	Laptop or desktop computer
Processor	Intel Core i5 or equivalent
Memory	8 GB RAM or more
Storage	At least 2 GB free space for development and datasets
Internet	Required for package installation; not required for local use after setup

2.2 Software Requirements

Software	Purpose
Operating System	Windows 10/11
Programming Language	Python 3.x
Data Analysis	pandas and NumPy
Machine Learning	scikit-learn
Visualization	Plotly for interactive charts; optional matplotlib/seaborn for offline EDA
Dashboard	Streamlit
Editor	Visual Studio Code or Jupyter Notebook
Version Control	Git; optional GitHub
Data Input	CSV real estate price dataset (for example, a suitable Kaggle source)

2.3 Development Tools and Frameworks

The analysis and model building are carried out in Python using pandas, NumPy and scikit-learn. Streamlit is used to turn the analysis into an interactive web dashboard with minimal front-end code, and Plotly provides zoomable, hoverable charts. Git supports source control and reproducible development.

CHAPTER 3
TECHNOLOGY STACK

3.1 Programming Languages

Python is used for data loading, preprocessing, statistical analysis, model training and the dashboard. SQL or plain CSV files may be used for storing the dataset.

3.2 Data Processing Libraries

pandas supports CSV loading, missing-value checks, grouping, filtering and descriptive statistics. NumPy provides numerical operations. Together they compute locality averages, price-per-square-foot values, correlations and other summary metrics.

3.3 Analytics and Machine Learning

scikit-learn is used for train/test splitting, categorical encoding, feature scaling and regression models. Linear Regression is the baseline; Random Forest Regression and Gradient Boosting Regression capture non-linear relationships. Models are compared using R^2, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) on a held-out test set. No accuracy is claimed in this report until the experiments are run.

3.4 Visualization

Plotly renders histograms of price, box plots by locality, scatter plots of area versus price, bar charts of average price per square foot and correlation heatmaps. Interactive hover and zoom let users examine individual data points.

3.5 Dashboard and Version Control

Streamlit organises the application into pages and sidebar filters, re-running calculations whenever the user changes an input. Git tracks changes to code, notebooks and documentation.

3.6 APIs or External Services

The current design uses a static dataset and does not depend on a live property API. Kaggle or an open government data portal may be used as the data source; the exact dataset name, author and license must be documented once selected. Live listing integration is not confirmed.

CHAPTER 4
SYSTEM ARCHITECTURE

4.1 High-Level Architecture

Layer	Responsibility	Input	Output
Data layer	Store and load property records	CSV files	Structured property data
Preprocessing & EDA	Clean, validate and analyze	Raw records	Clean data, statistics and summaries
Model layer	Train and evaluate regression models	Cleaned features	Trained model and metrics
Dashboard layer	Display filters, charts and estimates	User inputs, model output	Interactive pages

4.2 Description of Each Layer

Data Layer

Holds attributes such as location, area in square feet, bedrooms, bathrooms, age of property, amenities and price. Any personal information such as owner names or contact numbers should be excluded.

Preprocessing and EDA Layer

Checks missing and duplicate values, handles outliers, validates numeric ranges, derives price per square foot and calculates descriptive statistics and correlations.

Model Layer

Encodes categorical features, splits the data into training and test sets, trains the candidate regression models, compares their metrics and stores the selected model for the dashboard.

Dashboard Layer

Displays KPI cards, locality comparisons, charts and the price estimation form. Loading, empty-result and error states should be handled clearly.

4.3 Deployment Architecture

For local development the Streamlit application runs on localhost (port 8501 by default) and reads the dataset and saved model from the project folder. A public deployment would require hosting, a managed data store and secure configuration. Production hosting is not confirmed.

CHAPTER 5
DESIGN

5.1 Data Design

A real estate price dataset is the starting point. The exact fields depend on the dataset chosen. The following is the target schema; absent columns must be collected or derived before the related feature is enabled.

Field	Description
property_id	Unique identifier for each listing
location	City, locality or neighbourhood
property_type	Apartment, independent house, villa, plot, etc.
area_sqft	Built-up or carpet area in square feet
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms
age_years	Age of the property, if available
amenities	Parking, lift, security and other facilities, if available
price	Listing or sale price (target variable)
price_per_sqft	Derived: price divided by area

5.2 Data Quality and Preparation Design

The preprocessing workflow verifies column names and data types, removes duplicate listings, identifies missing values and checks that numeric fields fall in realistic ranges. Missing values should not be silently replaced by zero; any imputation or removal strategy must be documented before final statistics are calculated. Extreme outliers, such as unrealistic prices or areas, should be reviewed before being removed.

5.3 Dashboard Wireframes / Mock-ups

- Overview Page: KPI cards (number of listings, average price, median price per sq. ft.) and price distribution.
- Explore Page: sidebar filters for location, budget, area and bedrooms with a results table.
- Compare Page: locality-wise price comparison using bar and box plots.
- Analytics Page: area-versus-price scatter plot and correlation heatmap.
- Estimator Page: input form for property details and an indicative estimated price.

[Insert screenshots from the actual working Real Estate Price Exploration Tool here.]

5.4 Database Schema

If a database is used instead of CSV files, a suggested design is Locations(location_id, city, locality), Properties(property_id, location_id, property_type, area_sqft, bedrooms, bathrooms, age_years, price) and Amenities(property_id, amenity). This is a design proposal, not a verified implemented schema, and should be revised to match the final storage method.

CHAPTER 6
IMPLEMENTATION

6.1 Module-Wise Implementation

Module	Purpose	Input → Output
Data Loading & Cleaning	Load, validate and clean property data	CSV → clean DataFrame
EDA	Summarize and visualize patterns	Clean data → statistics and charts
Model Training	Train and compare regression models	Features → model and metrics
Price Estimator	Predict price from user inputs	Property details → estimated price
Dashboard	Present filters, charts and estimates	User actions → interactive pages

6.2 Data Preprocessing

- Load the selected CSV file and inspect its columns and data types.
- Standardize column names and convert numeric fields such as area and price.
- Report missing values and duplicates, then apply documented cleaning rules.
- Derive price per square foot and, where needed, convert units such as lakhs or crores to a single unit.
- Encode categorical variables such as location and property type (for example, one-hot encoding).

6.3 Exploratory Data Analysis

EDA examines the distribution of price and area, compares average price per square foot across localities, studies how bedrooms and property age relate to price, and computes a correlation matrix for numeric variables. Correlation is reported as an association, not as proof of causation.

6.4 Model Building

Model	Reason for Selection	Output
Linear Regression	Simple, interpretable baseline	Coefficients and predicted price
Random Forest Regression	Handles non-linear relationships and mixed features	Predicted price and feature importance
Gradient Boosting Regression	Often accurate on tabular data	Predicted price and feature importance

The data is split into training and test sets, and models are compared using R^2, MAE and RMSE. Cross-validation may be used for more reliable comparison. The best model is saved and used by the estimator page.

6.5 Dashboard Integration

The Streamlit application loads the cleaned dataset and the saved model, applies the filters chosen in the sidebar, and re-draws charts and tables accordingly. The estimator form passes user inputs through the same preprocessing steps used in training before calling the model. Estimates are displayed with a clear note that they are indicative only.

CHAPTER 7
FEATURES

7.1 Main Features

- Summary cards for number of listings, average price, median price and average price per square foot.
- Interactive filters for location, price range, area, bedrooms and property type.
- Locality comparison charts and price-per-square-foot rankings.
- Area-versus-price scatter plot and correlation heatmap.
- EDA tables for descriptive statistics and missing values.
- Indicative price estimation from property details.
- Downloadable filtered results as a CSV file.

7.2 User Workflow

1. The user opens the dashboard in a web browser.
2. The user applies filters for location, budget and property attributes.
3. The application updates the summary cards, charts and tables for the filtered data.
4. The user compares localities or inspects relationships between features and price.
5. The user enters property details in the estimator to obtain an indicative price.

7.3 Intended Use and Disclaimer

The tool is for education, research and preliminary planning. Estimates depend on the quality and coverage of the dataset and should not be used as the sole basis for buying, selling, lending or investment decisions.

CHAPTER 8
TESTING

8.1 Validation Approach

Testing covers CSV loading, data cleaning, calculation of statistics, filter behaviour, model prediction, dashboard rendering and error handling. The cases below are proposed acceptance checks; they are not recorded as passed until executed against the final implementation.

8.2 Test Cases and Expected Results

Test ID	Test Description	Expected Result	Status
TC-01	Load a valid CSV file	Data loaded and summaries displayed	To verify
TC-02	Load a CSV missing a required column	Clear validation error	To verify
TC-03	Apply location and budget filters	Only matching properties shown	To verify
TC-04	Dataset contains missing values	Handled according to documented policy	To verify
TC-05	Enter valid property details in the estimator	Estimated price displayed	To verify
TC-06	Enter invalid or empty estimator inputs	Readable error message; no crash	To verify
TC-07	Filters return no results	Friendly empty-state message	To verify

8.3 Model Evaluation Metrics

Model quality is assessed on a held-out test set using R^2 (proportion of variance explained), Mean Absolute Error and Root Mean Squared Error. Dashboard aggregates such as mean price and listing counts should also be checked against independently calculated values from the same CSV. Metric values must be taken from actual runs and reported with the dataset size and split used.

8.4 Performance Testing

Measure dashboard load time, filter response time and model prediction time on the target development machine, and record the dataset size and hardware used. No benchmark figures are asserted here.

CHAPTER 9
DEPLOYMENT

9.1 Development Environment

The project can be developed on Windows using Visual Studio Code or Jupyter Notebook, Python and Git. A virtual environment is recommended so that package versions remain consistent.

9.2 Dashboard Execution

Create and activate a virtual environment, install the required packages (for example, pandas, numpy, scikit-learn, plotly and streamlit) and start the dashboard with the command streamlit run app.py, assuming the entry file is named app.py. The actual repository layout should be checked before submission.

9.3 Environment Configuration

Keep a requirements.txt file listing package versions, document the dataset path and the saved model file, and keep any credentials outside source control. If the dashboard is hosted publicly, use HTTPS and avoid publishing personal data contained in listings.

CHAPTER 10
CHALLENGES AND LIMITATIONS

10.1 Challenges Faced

- Property datasets differ in columns, units and naming of localities.
- Missing, duplicate or inconsistent listing data can distort statistics and predictions.
- Outliers such as unusually high prices can mislead models if not handled carefully.
- Prices change over time, so older data may not reflect current market conditions.
- Encoding many locality names can create very large feature sets.

10.2 Solutions and Mitigations

The design uses explicit validation, documented cleaning rules, consistent unit conversion, outlier review and comparison of several models. Rare localities can be grouped, and the date of the data should be recorded so that users know how current it is.

10.3 Current Limitations

- Available features depend on the columns in the selected dataset.
- The dataset may not represent every city, locality or property segment.
- Predicted prices are estimates and not official valuations.
- Live listing integration and production-grade hosting are not confirmed.
- The report does not contain verified accuracy figures or final screenshots.

CHAPTER 11
FUTURE ENHANCEMENTS

11.1 Planned Improvements

- Connect to live or regularly updated listing data sources.
- Add map-based visualization of prices by locality.
- Include features such as distance to metro, schools and hospitals.
- Add price trend forecasting using time-series methods.
- Provide downloadable PDF market reports for selected localities.
- Add user accounts to save searches and compare shortlisted properties.

11.2 Possible Applications

- Home buyers and tenants researching localities.
- Real estate agencies preparing market summaries.
- Banks and analysts studying regional price patterns.
- Academic projects on housing data and regression modelling.

CHAPTER 12
CONCLUSION

12.1 Summary of the Project

Real Estate Price Exploration Tool brings together data cleaning, exploratory analysis, interactive visualization and machine learning based price estimation in a single dashboard concept. It aims to make property price data easier to explore and to support better informed decisions.

12.2 Project Outcomes

The project scope includes interactive filtering, locality comparison, descriptive statistics, correlation analysis and an indicative price estimator. Features and results must be verified against the final running application before being described as completed.

12.3 Skills Learned During Development

- Data cleaning and exploratory data analysis using Python.
- Building and comparing regression models with scikit-learn.
- Creating interactive visualizations and dashboards.
- Version control, testing, debugging and technical documentation.
- Working as a team on a data-driven project.

CHAPTER 13
REFERENCES

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[4] Plotly Python Documentation, https://plotly.com/python/

[5] Streamlit Documentation, https://docs.streamlit.io/

[6] Kaggle Datasets, https://www.kaggle.com/datasets

[7] Git Documentation, https://git-scm.com/doc

Final dataset citation: [Insert exact dataset title, author, URL, license and access date after selection.]

Project repository: [Insert the actual Real Estate Price Exploration Tool repository URL, if published.]

CHAPTER 14
APPENDICES

14.1 Data Analysis and Model Pseudocode

```
data = pandas.read_csv("real_estate_prices.csv")
validate_required_columns(data)
data = clean_records_using_documented_rules(data)
data["price_per_sqft"] = data["price"] / data["area_sqft"]
locality_summary = data.groupby("location")["price_per_sqft"].mean()
missing = data.isna().sum()
X, y = encode_features(data), data["price"]
X_train, X_test, y_train, y_test = train_test_split(X, y)
model = GradientBoostingRegressor().fit(X_train, y_train)
metrics = evaluate(model, X_test, y_test)  # R2, MAE, RMSE
```

The pseudocode assumes columns named location, area_sqft and price. Adjust it to the final dataset schema.

14.2 Screenshot Checklist

- Overview page with KPI cards and price distribution.
- Filter sidebar and filtered results table.
- Locality comparison charts.
- Area-versus-price scatter plot and correlation heatmap.
- EDA statistics and missing-value tables.
- Price estimator form with an example result.
- Terminal showing successful local launch of the dashboard.

14.3 Installation / Setup Information

- Record the final Python and package versions.
- Provide requirements.txt for the project.
- Identify the actual dataset file and its source.
- Document the exact launch command and local URL.
- Insert screenshots and results captured from the working application.

14.4 Important Submission Note

This report adapts the chapter structure of the supplied reference document to the Real Estate Price Exploration Tool project. Proposed features, schema and test cases must be checked against the final code. Replace all placeholders (guide and HOD names, screenshots, dataset citation) and any unverified claims before academic submission.